

Amazon Product Recommendation System: Approach Document

Case Study Overview

The primary objective of this case study is to design and evaluate a **Product Recommendation System** for an e-commerce platform like Amazon. You will work towards building a **Hybrid Recommendation Engine**, combining **Content-Based Filtering** and **Collaborative Filtering** approaches.

You should approach this case as a **Data Scientist working in a product analytics team**, responsible for converting raw product and user interaction data into meaningful insights and reliable recommendations that can improve user experience and business outcomes.

This document provides **hints** on how to approach each section of the case study. It does not reveal answers or exact steps. The final decisions and implementation choices are intentionally left to you.

Dataset Overview

You are provided with a dataset (amazon.csv) that contains detailed information about products listed on the platform along with user feedback and engagement signals.

Key Data Fields to Explore

Product Information

- Product identifiers and names
- Product categories and descriptions

Pricing & Discounts

- Actual price and discounted price
- Discount percentage

User Interaction Data

- User ratings and number of ratings
- User identifiers and review text

Understanding how these fields relate to one another is critical before moving into analysis or modeling.

Section A: Data Understanding & Cleaning

Goal

Prepare the raw dataset for analysis by fixing formatting issues, handling missing or incorrect values, and creating useful features.

1. Data Inspection & Summary

Before starting data cleaning, focus on understanding the overall structure and scale of the data.

Hint:

- Identify the number of unique users and products present in the dataset. This helps you understand how dense or sparse the data is, which becomes important later for recommendation modeling.

Hint:

- Look at how many interactions exist per user and per product at a high level. This gives an early sense of whether most users interact with many products or only a few.

Hint:

- Study the product category column. Categories are often stored in a hierarchical format (for example: **Electronics | Mobiles | Accessories**).

Hint:

- Consider extracting only the top-level or primary category to make aggregation and analysis easier.
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2. Data Cleaning & Preprocessing

The raw data may contain formatting issues that prevent direct analysis.

Hint:

- While cleaning, think about whether an issue is a true data error or a realistic business scenario (for example, missing ratings for newly listed products).

Price Columns

- Pricing-related columns often include currency symbols or separators that make them unusable for calculations.

Hint:

- Clean these columns so they can be treated as numeric values.

Ratings

- Review the ratings column carefully to ensure it contains valid numeric values.

Hint:

- Handle missing, incorrect, or inconsistent rating values in a way that makes sense from a business point of view.

Category Parsing

- Product categories are usually stored as a single text field with multiple levels.

Hint:

- Split this information into simpler components (such as main category and sub-category) to support better analysis later.

The outcome of this section should be a **clean, consistent dataset** ready for exploration and modeling.

Section B: Exploratory Data Analysis (EDA)

Goal

Analyze the data to understand product performance, user engagement, and overall buying patterns.

Hints

- Compare multiple signals together (for example, ratings along with review counts) instead of looking at any one metric in isolation.

- Look for patterns that repeat across categories rather than focusing only on individual products.
- Pay attention to outliers, as they may represent either data issues or strong business opportunities.

Insights

- Focus on insights that can help improve product discovery, pricing strategy, or customer experience.
 - Aim to generate observations that a business or product team could realistically act upon.
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Section C: Content-Based Recommendation Approach

Goal

Build a recommendation approach based on **product attributes**, without relying on user behavior history.

Hints

- Treat each product as a collection of descriptive features such as name, description, category, price, and discount.
- Convert these features into a structured form that allows you to measure similarity between products.
- Think about similarity from a customer's point of view, not just a technical one.

Evaluation

- Check whether the recommended products are relevant and logically related.
 - Ensure that recommendations are not repetitive and offer some variety.
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Section D: Collaborative Filtering Approach

Goal

Use **user behavior data** to generate personalized product recommendations.

Hints

- Represent user-product interactions in a structured format suitable for comparison.
- Decide whether to focus on user-based or item-based similarity, and explain your reasoning.
- Be aware of challenges such as sparse data and users with very limited history.

Evaluation

- Review whether the recommendations reflect known user preferences.
 - Identify situations where collaborative filtering may struggle to perform well.
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Section E: Hybrid Recommendation Strategy

Goal

Combine content-based and collaborative filtering approaches to create a **more balanced recommendation system**.

Hints

- Think about why a single recommendation approach may fail in certain situations and how combining methods can reduce those gaps.
- Compare approaches not just on accuracy, but also on coverage, relevance, and reliability across different user types.
- Evaluate the hybrid approach separately for new users, returning users, and new products.

Also consider real-world factors such as product popularity, recent activity, and availability.

Section F: Business Strategy & Deployment

Goal

Connect your analytical and modeling work to real-world business use cases.

Hints

- Decide which recommendation approaches are better suited for new users versus returning users.

- Think about strategies for recommending products that have very little or no user feedback.
- Outline how this system could be deployed, monitored, and updated in a production environment.

Success Measurement

- Identify key metrics that reflect user engagement and business impact.
 - Explain why these metrics are important for evaluating a recommendation system.
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Final Notes

- There is no single correct solution for this case study.
- Clear thinking, logical structure, and justification of decisions are more important than complex techniques.
- Treat this exercise as a real-world simulation of how recommendation systems are built in industry.